

Project SHIELD 2.0

We teach teenagers about scams by letting them get scammed in a simulated environment. The version we run now (SHIELD 1.0) works well enough to prove the idea *somewhat*. Here's some context for the next one:

Subject	Scams and social engineering, for young people up to 25
In this pack	Where we are, what went wrong, and what you can do

Who we are

SCOGA, the Singapore Cybersports and Online Gaming Association, is a non-profit that has spent over a decade using games as the way in to subjects people find hard to talk about. Digital safety, careers, loneliness in older adults, drug abuse prevention. We run programmes in secondary schools, ITEs, polytechnics, universities, community centres and senior activity centres.

Esports Academy Pte Ltd is the education arm. It writes the curriculum, trains the facilitators and delivers the workshops. Nearly everything we build sits on game-based and experiential learning: students play, decide, get it wrong safely, and work out why afterwards. That's the method behind SHIELD and behind most of what we run.

We are not a tech company. We know what needs to happen in a classroom and we are confined by our non-tech background most of the time.

People learn by doing, not by being told about it. If a workshop ends with students able to recite the right answer but no better at acting on it, it didn't work, however good the content was.

What SHIELD 1.0 is, and why we built it

Every digital safety programme we'd seen taught the same way. An adult lists the warning signs and asks the class to remember them. Students pass the quiz easily. They'll tell you a stranger asking for photos is a red flag. Yet when it happens for real, no one really knows what to do.

Knowing the answer and doing the right thing under pressure are separate skills, and only one of them was being taught. So we built the other one.

SHIELD 1.0 drops a student into a live conversation with an AI

persona who has an agenda. A stranger who's a bit too interested. A classmate spreading something cruel. A seller with a deal that's slightly too good. The student replies in their own words, in real time, without knowing where it's heading. Afterwards a facilitator debriefs the room on what just happened and why.

It runs as a facilitator-led workshop of about 1.5 hours, in a classroom, and we've delivered it to secondary school students across Singapore. It's built on Capybara, a third-party AI persona platform that we bent into a teaching tool. Modules cover unwanted contact, cyberbullying, misinformation and scams.

SHIELD 1.0 and its issues.

1. The platform

Capybara has proved the idea works. We got two years of real classroom evidence out of it without building anything ourselves.

The limits are real though. It isn't realistic enough to hold a student's belief, and the personas drift out of character. It's slow, it hits conversation limits partway through a scenario, and it looks like nothing a fifteen year old has ever chosen to use.

2. The conversation, which is the real problem

SHIELD 1.0 uses AI but...

The conversations are fixed and stiff. The persona has a script it's determined to finish, so it steers back to its own lines whatever the student says. Go off point and it either ignores you or produces something that makes no sense. It behaves nothing like the AI these students already use every day, which is fluid, responsive and occasionally surprising.

Students kinda clock it quickly, and the feedback is consistent:

It doesn't talk like a real person.

The comment we get most often. Almost every single student.

You can tell it's fake, so you stop trying.

Once the illusion goes, the engagement goes with it. Performing for the sake of it.

Nobody would actually message me like that.

The scenarios read as though they were written by adults, because they were.

That matters more than it might sound. The method only works if a student is drawn in far enough to make a real decision under real uncertainty. If they can see the scaffolding from the first message they aren't deciding anything, **they're performing the correct answer for the adult at the front of the room. Which**

is the thing we built SHIELD to get away from in the first place.

If you take one thing from this document

Make it feel real. Most of what follows is negotiable, solvable, or our problem rather than yours. Realism is the part we can't figure out, and it's what we'd most like to see somebody crack.

The SHIELD 2.0 brief

SHIELD 2.0 is funded by IMDA's Digital for Life fund and narrows to a single topic: scams and social engineering. Not digital safety generally.

The aim is to get children and young people up to 25 able to spot a scam as it's happening and know what to do about it. They talk to an AI persona inside a simulated scam and decide, as it unfolds, whether to keep talking, hand over information, pay, or walk away. The scenario reacts to what they choose. We've committed to four of them, all pulled from where young people actually spend their time online.

We deliver this with MOE schools, institutes of higher learning and community partners, in classrooms and at public events.

The four scenarios

Scenario	What it looks like	The tactic being taught
Job and side-hustle	Easy money, flexible hours, no experience needed. Then there's a fee to pay, or a bank account to lend, or documents to hand over.	Trust in a platform that looks official, plus real financial pressure on young people who want their own money.
In-game item and account trade	A skin, an account or a rare item at a price that's good without being impossible. Payment goes one way and nothing comes back.	Familiarity inside a gaming community. Everyone trades like this, so the request feels normal.
Romance and friendship grooming	Someone attentive and interested who's always there. The ask comes weeks in, at a point where saying no would feel like letting down a friend.	A relationship built deliberately over time, then spent.
Friend impersonation	A message from an account that is your friend's, in your friend's voice, asking for something small and urgent.	Impersonating a known contact. The trust is real, the person using it isn't.

Fixed scaffold, fluid conversation

This is the design principle underneath SHIELD 2.0 currently.

Earlier, there are two obvious ways to build this and both have their issues.

The first is to script it. Write the whole conversation, branch it, let the student choose from options. This allows complete control over the conversation and pedagogically it's neat, but it's also very stale for students. It makes the conversation feel stiff and fake.

The second is to let the model run. Prompt an LLM to play a scammer and let it improvise. It worked wonderfully in a demo with perfect students who talk the way we predicted they would. Then in an actual classroom it wanders off topic, gets talked out of character by very creative students, misses the learning objective entirely, or says something truly unhinged and not meant for public eyes.

What we want sits between them. The structure is fixed and written by us. The conversation inside it is generated and free.

Fixed: we write it, it can't drift	Fluid: the model handles it, freely
• The scam's arc, from the setup through the hook and the pressure to the ask • Which manipulation tactic each beat is teaching • The decision points the student has to be given • Which choices are safe and which aren't • What the debrief says at the end • The hard limits on what the persona will and won't say	• The persona's name, voice, backstory and typing style • How it gets from one beat to the next • Its reply to whatever the student actually typed • Pushing harder or backing off depending on how the student is doing • Which game, platform or job the scam is dressed in • How long it lingers before making the ask

So a student can type anything and get a real answer, but the conversation always passes through the beats we care about and always ends somewhere we chose. The model has a lot of freedom in how it gets there and none at all in where it goes.

Someone trying to break it should find that fun and get nowhere. Someone playing along should never see the rails.

We're giving you the principle, not the mechanism. How you actually hold a scaffold on a generative model is wide open. The short version: we need some assurance the AI won't behave too unpredictably in a classroom.

The playing field for SHIELD 1.0 so far

The classroom

- Thirty to forty students, one facilitator, no IT support.
- The simulation is only part of the session. There's a briefing before and a debrief after.

The technology

- Browser only. No app, no install, no app store.
- Students use their school PLDs, which are usually iPads, thus browser.

The students

- They're minors, which governs everything else in this list.
- Collect as close to nothing as you can. Our preference is no

accounts at all. If you can make it work with zero personal data, would be best.

- Never ask a student for real names, contact details, or anything about their own experience of being scammed or approached. Some of them have been, and a workshop isn't the place for that to come out.

Hard no's

Things we won't take forward, however well built.

- Open chat that can wander. If a student can talk the persona onto another subject, it's out. Keep the fluidity and lose the freedom to leave the topic.
- Anything that teaches usable scam technique. The student is always the target and never the operator. No build your own scam, no playing as the scammer, no phishing kit however educational the framing. We're teaching recognition, not method.

Everything not on that list is fair game, including things we'd never have come up with ourselves.

Problems we haven't solved

These are real and open. We don't have answers to them. A decent attempt at either would interest us more than a polished build that avoids both.

Time compression

A romance or grooming scam works because it takes weeks. The attentiveness, the daily messages, the slow accumulation of trust. That isn't the run up to the scam, it's the mechanism itself. The ask only lands because of everything that came before it.

We have about twenty minutes. Compress the timeline and you lose the thing you're trying to teach, because the student watches a friendly stranger ask for money on day one and concludes, correctly, that they'd never fall for it. Which is the opposite of the lesson.

We don't know how to make weeks feel like weeks in a classroom. Someone might.

Realism against safety

These two pull against each other directly. The more convincing the persona, the better it teaches, and the more careful we have to be about what it's capable of saying to a child. Every step towards realism is a step towards something we might struggle

to defend to a parent. Where that line sits, and how you hold it without flattening the experience, is something we haven't settled.

Getting into SHIELD 1.0

If you would like to try out SHIELD 1.0, here are some logins you can use.

Your login details

GenAI Platform	https://www.scoga.org/shield Login Details: Shield[NUMBER]@esportsacademy.sg Smart1234!	[NUMBER] runs from 001 to 099
Example	Login: shield001@esportsacademy.sg Password: Smart1234!	

Two things you can do:

1. Go through a scenario the way a student would.
2. Look through the backend prompts.

What happens to what you build

We're partway through designing SHIELD 2.0 and we really want to see the ideas that can come out of this hackathon.

We may eventually choose one team's idea and take it forward to shape the real product. SHIELD 2.0 then runs for two years, funded and built properly, delivered in secondary schools, ITEs, polytechnics, universities and community events across Singapore.

The team can get the opportunity to be pulled into the actual build conversation, as contributors or in the vendor discussions.

On the day

What we're hoping to see by the end of the day, roughly in order of how much we care:

1. Something that feels real, that a fifteen year old would be drawn into rather than perform for.
2. Something with a spine, where the conversation is free but the lesson still lands.
3. Something that survives a real classroom: thirty-nine other devices, bad wifi, two minutes to get everyone started and no adult free to hold its hand.

